

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q12-0140

Buck Hydro 138 kV
June 24, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0140. The Interconnection Customer proposes a 204.7 MW solar photovoltaic facility interconnecting at the Buck Hydro 138 kV substation in the Virginia area. The request was received on June 26, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 94 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$57,080,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Silver City / Tricou 138 KV: 132.3% loading (245 MVA) following the outage of the 4 Ac Sta / Ave 52 138 KV L; DFAX = 100.0%.

Reserver Intl / Tricou 138 KV L: 129.6% loading (236 MVA) following the outage of the 4 Ac Station / Ave 52 138 KV line; DFAX = 100.0%.

Copco 2 / Delta Power Substation 138 KV: 128.5% loading (224 MVA) following the outage of the Ave 52 / 4 Ac Sta 138 KV line; DFAX = 100.0%.

The 4 Ac Station / Ave 52 138 KV line is overloaded to 128.2% of its 252 MVA emergency rating for the loss of the Tricou / Silver City 138 KV. The project contributes 100.0% DFAX (204.7 MW) to this overload.

For the loss of 4 Ac Station / Ave 52 138 KV line, the Reserver Intl / Copco 2 138 KV loads to 127.6% of its 229 MVA rating. Project DFAX: 100.0%.

Buck Hydro / 4 Ac Sta 138 KV: 127.5% loading (247 MVA) following the outage of the Silver City / Tricou 138 KV Line; DFAX = 100.0%.

Jacksons Ferry / Ave 52 138 KV Line: 124.5% loading (245 MVA) following the outage of the Tricou / Silver City 138 KV; DFAX = 100.0%.

For the loss of Tricou / Silver City 138 KV, the Jacksons Ferry / No Troy 138 KV loads to 124.2% of its 241 MVA rating. Project DFAX: 100.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Silver City / Tricou 138 KV	Rebuild line with higher-capacity conductor	\$3,155,000	100% (\$3,155,000)
Reserver Intl / Tricou 138 KV L	Rebuild line with higher-capacity conductor	\$6,606,000	100% (\$6,606,000)
Copco 2 / Delta Power Substation 138 KV	Rebuild line with higher-capacity conductor	\$9,261,000	100% (\$9,261,000)
4 Ac Sation / Ave 52 138 KV line	Rebuild line with higher-capacity conductor	\$14,967,000	100% (\$14,967,000)
Reserver Intl / Copco 2 138 KV	Rebuild line with higher-capacity conductor	\$8,875,000	100% (\$8,875,000)
Buck ydro / 4 Ac Sta 138 KV	Rebuild line with higher-capacity conductor	\$1,729,000	100% (\$1,729,000)
Ave 52 Substation / Jacksons Ferry 138 KV Line	Rebuild line with higher-capacity conductor	\$6,359,000	100% (\$6,359,000)
Jacksons Ferry / No Troy 138 KV	Rebuild line with higher-capacity conductor	\$6,128,000	100% (\$6,128,000)
	Total allocated to project		\$57,080,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.